

Nazmul Alam Diptu

✉ diptunazmulalam@gmail.com | 🌐 diptu.github.io | 🎓 Google Scholar | 🔗 linkedin.com/in/diptu | 📄 github.com/diptu

📍 Dhaka, Bangladesh | 🎯 Targeting Ph.D. / M.S. Positions (GRA/GTA)

RESEARCH SUMMARY

Computer Science researcher and ML engineer focused on ****Edge AI, TinyML, and Low-Latency Time-Series Systems****. Experienced in building resource-aware machine learning pipelines under severe compute, memory, and energy constraints. Seeking ****Ph.D. or Research-Track M.S. positions (GRA/GTA)**** in ML Systems and Efficient Computing.

FEATURED RESEARCH SYSTEMS ARTIFACT

- EcoLens: High-Throughput Energy Intelligence Platform** | *ML Systems, TinyML, Pipeline Optimization* 2025 – 2026
- Engineered an end-to-end, real-time time-series forecasting system for electricity demand and dynamic carbon intensity across Australian electricity grid markets.
 - Optimized inference runtimes to achieve **sub-100ms response times** under continuous streaming workloads while maintaining low memory footprints.
 - Implemented an asynchronous ETL/ELT pipeline (FastAPI, Docker, PostgreSQL) with automated drift detection and backtesting loops to handle continuous sensor feeds.

PUBLICATIONS

- End-to-End OCR via Synthetic Dataset Generation** (*IJCCI 2020, Springer*) Md. Shopon, **N. A. Diptu**, N. Mohammed
– Trained deep learning OCR models on synthetic noisy images, improving robustness in document recognition | doi:10.1007/978-981-15-3607-6_41
- Population Estimation of Rohingya Refugees via ANN** (*VJCS 2019, World Scientific*) N. Ahmed, **N. A. Diptu**, et al.
– Modeled refugee population dynamics using Artificial Neural Networks on satellite and census data | doi:10.1142/S2196888819500246

EXPERIENCE

- North South University** May 2019 – Jul. 2022, Feb. 2025 – Present
Lab Instructor Dhaka, Bangladesh
- Instructed 1000+ undergraduates in **C, Java, Data Structures, Database Systems, and Web Technologies**.
 - Mentored students in core CS theory, algorithmic complexity, and led lab coursework evaluation framework design.
 - *Note: Service paused Aug 2022 – Jan 2025 due to full-time software/ML industry engineering commitments.*
- Me-Solshare Ltd** Nov. 2019 – Aug. 2021
Junior Data Scientist Dhaka, Bangladesh
- Developed CNN-based time-series forecasting models achieving **** 95% accuracy**** for solar grid energy consumption.
- Power Ledger Pty Ltd** Oct. 2022 – Feb. 2024
Product Analyst Remote
- Managed technical data analysis, platform metrics, and system quality assurance for decentralized energy trading.

EDUCATION

- North South University** Jan. 2015 – Dec. 2018
B.Sc. in Computer Science and Engineering CGPA: 3.62 / 4.00
- **Honors:** Awarded 25% Merit Scholarship based on academic performance.
 - **Relevant Coursework:** Data Structures, Algorithms, Databases, Machine Learning Foundations, Probability & Statistics.

SKILLS

ML Frameworks & Deployment : PyTorch, TensorFlow, ONNX Runtime, TFLite, scikit-learn
Languages : Python, C/C++, Java, JavaScript/TypeScript, SQL
MLOps & Systems Tools : FastAPI, Docker, PostgreSQL, Apache Airflow, dbt, Git, Linux, $\LaTeX$